Sai Chaitanya Susarla

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Github profile

Postdoctoral fellow in the School of Mathematics at the **University of Southampton, UK**. My current focus is on advancing profile domain timing methods using Machine learning techniques.

Personal Data

Date of Birth : 17th January 1999
Place of Birth : India
Nationality : Indian
Current Residence : Southampton, United Kingdom
ORCID ID : 0000-0003-4332-8201

Work Experience

- **Postdoctoral fellow: School of Mathematics** — **Southampton, UK**
  University of Southampton — *May 2026 - Present*
  Project title: Profile Domain timing using machine learning techniques.
- **Postdoctoral fellow: Department of Physics** — **Egham, UK**
  Royal Holloway, University of London — *October 2025 - May 2026*
  Project title: Inferring the timing properties of neutron stars with hierarchical profile-domain timing.

Education

- **Doctoral Studies: Physics and Astronomy** — **Galway, Ireland**
  University of Galway — *2021–2025*
  Thesis title: Examining ecliptic pulsars using the low frequency telescopes
- **BS-MS Dual Degree: Physics Major, Maths Minor** — **Thiruvananthapuram, India**
  Indian Institute of Science Education and Research, Thiruvananthapuram, — *2016–2021*
  8.37/10 CGPA
- **Higher Secondary Certificate – 89.00%** — **Pune, India**
  Kendriya Vidyalaya Southern Command, — *2014–2016*
  In top 1% of Central Board of Secondary Education

Scholarships/Fellowships

- January 2023 - *APS FIP Distinguished Student Award* by the American Physical Society, Forum for International Physics.
- August 2021 - *College of Science and Engineering Fellowship* by the College of Science and Engineering at the University of Galway.
- August 2021 - *International Merit Scholarship* by the University of Galway which covers the fees for Graduate studies.
- June 2017 - *Visiting Students Programme* at National Centre for Radio Astrophysics–Tata Institute of Fundamental Research.
- September 2016 - *Inspire SHE Scholarship* by Department of Science and Technology, Government of India.

Talks & Posters

Invited talks

- **October 2026 - Scintillometry 2026 workshop, Cornell University, USA** - Title– If you can simulate, can you measure it?
- **Feb 2026 - CPPA Seminar, Royal Holloway University of London, London, UK** - Title– Exploring the ecliptic pulsars using the low frequency telescopes
- **May 2025 - SHAO Seminar, Shanghai Astronomical Observatory, Shanghai, China** - Title– Radio eyes on the sky: Using pulsars to probe the Sun and detect Gravitational ripples.

Contributed talks

- **Sept 2026 - SPINS UK 2026, Bath, United Kingdom** - Simulation-based inference of profile evolution for high precision pulsar timing.
- **June 2026 - International Pulsar Timing Array meeting 2026, Drakensburg, South Africa** - Talk on Modelling frequency evolution using Simulation-based inference techniques.
- **May 2025 - European Pulsar Timing Array spring meeting 2025, Bonn** - Talk on Solar wind model comparison and use cases for EPTA-LOFAR data combination.
- **Oct 2024 - Physics Seminar, University of Galway** - Talk on Pulsar observations using I-LOFAR.
- **June 2024 - International Pulsar Timing Array meeting 2024, Sexten, Italy** - Talk on Exploring the time variability of solar wind and comparison with recent models for pulsars using LoFAR dataset.
- **May 2024 - Departmental presentation, University of Galway** - Talk on Effects on Solar wind on LOFAR pulsar data.
- **March 2024 - European Pulsar Timing Array Spring meeting 2024, Saumur, France** - Talk on Exploring the time variability of solar wind using pulsars from the LoFAR dataset.
- **November 2023 - European Pulsar Timing Array Fall meeting 2023, Oxford** - Talk on Exploring the effects of solar wind on pulsars.
- **August 2023 - Irish National Astronomical Meeting 2023, Cork**- Talk on Solar Wind effects and how to disentangle them?
- **June 2023 - 15th Edoardo Amaldi Conference on Gravitational Waves 2023, Virtual** - Poster on Exploring the effects of the Solar Wind on pulsars using the LoFAR telescope
- **April 2023 - American Physical Society April Meeting 2023, Minnesota** - Talk on Exploring the effects of Solar Wind on pulsars using the LOFAR telescope.

Publications

All the statistics here are updated according to NASA/SAO Astrophysical Data system as of 30th June 2026.
All the publications can also be viewed in the ORCID profile (ID: 0000-0003-4332-8201) or alternatively in the ADS system: (link)
Google Scholar profile: (link)
h-index of refereed publications: 17
Total number of citations: 3619
First author refereed publications: 3 publications in A&A, 1 submission in MNRAS

List of publications since 2020 (Recent first):

[1] **Sai C. Susarla**, Gregory Ashton, A. Parthasarathy, K. De Man & J. P. W. Verbiest. "Simulation-based inference of profile evolution for high-precision pulsar timing". In: (2026). arXiv: 2610.31450 [astro-ph.IM]. URL: https://arxiv.org/abs/2610.31450.

[2] Caterina Tiburzi, Michael Lam, Daniel Reardon, Nataliya Porayko, Maaijke Mevius, Stella Koch Ocker, **Sai Susarla**, et al. "Exploring the Galactic plasma with pulsars in the SKA Era". In: *The Open Journal of Astrophysics* 8 (Dec. 2025), p. 54250. DOI: 10.33232/001c.154250. arXiv: 2512.16158 [astro-ph.HE].

[3] F. Iraci, A. Chalumeau, C. Tiburzi, J. P. W. Verbiest, A. Possenti, **S. C. Susarla**, M. A. Krishnakumar, et al. "Combining the second data release of the European Pulsar Timing Array with low-frequency pulsar data". In: *A&A* 704, A109 (Dec. 2025), A109. DOI: 10.1051/0004-6361/202555516. arXiv: 2510.04639 [astro-ph.HE].

[4] **Susarla, S. C.**, Ferguson, R., Johnson, O. A., Vincetti, L., McKenna, D. J., Keane, E. F., McCauley, P. J., et al. "RRAT-like behaviour of PSR B0656+14 observed with I-LOFAR". In: (2025). arXiv: 2507.04518 [astro-ph.HE]. URL: https://arxiv.org/abs/2507.04518.

[5] **S. C. Susarla**, O. A. Johnson, D. J. McKenna, E. F. Keane, P. J. McCauley, J. P. W. Verbiest, C. Tiburzi, et al. "Long-term timing results of ecliptic pulsars observed with I-LOFAR". In: *A&A* 698, A248 (June 2025), A248. DOI: 10.1051/0004-6361/202554226. arXiv: 2505.09549 [astro-ph.HE].

[6] N. K. Porayko et al. "Searches for signatures of ultralight axion dark matter in polarimetry data of the European Pulsar Timing Array". 2025. arXiv: 2412.02232.

[7] K Liu, A Parthasarathy, M Keith, C Tiburzi, **S C Susarla**, J Antoniadis, A Chalumeau, et al. "The impact on astrometry by solar-wind effect in pulsar timing". In: *Monthly Notices of the Royal Astronomical Society* (Dec. 2024), stae2727. ISSN: 0035-8711. DOI: 10.1093/mnras/stae2727. eprint: https://academic.oup.com/mnras/advance-article-pdf/doi/10.1093/mnras/stae2727/61075114/stae2727.pdf. URL: https://doi.org/10.1093/mnras/stae2727.

[8] F. Iraci, A. Chalumeau, C. Tiburzi, J. P. W. Verbiest, A. Possenti, G. M. Shaifullah, **S. C. Susarla**, et al. "Pulsar timing methods for evaluating dispersion measure time series". In: *arXiv e-prints*, arXiv:2410.22170 (Oct. 2024).
Accepted in A&A. arXiv:2410.22170. DOI: 10.48550/arXiv.2410.22170. arXiv: 2410.22170 [astro-ph.HE].

[9] **S. C. Susarla**, A. Chalumeau, C. Tiburzi, E. F. Keane, J. P. W. Verbiest, J. S. Hazboun, M. A. Krishnakumar, et al. "Exploring the time variability of the Solar Wind using LOFAR pulsar data". In: *arXiv e-prints*, arXiv:2409.09838 (Sept. 2024).
Accepted in the Sun & the Heliosphere in A&A. arXiv:2409.09838. DOI: 10.48550/arXiv.2409.09838. arXiv: 2409.09838 [astro-ph.SR].

[10] EPTA Collaboration et al. "The second data release from the European Pulsar Timing Array. IV. Implications for massive black holes, dark matter, and the early Universe". 2024. arXiv: 2306.16227.

[11] G. Agazie et al. "Comparing Recent Pulsar Timing Array Results on the Nanohertz Stochastic Gravitational-wave Background". 2024. arXiv: 2309.00693.

[12] Hippolyte Quelquejay Leclere et al. "Practical approaches to analyzing PTA data: Cosmic strings with six pulsars". 2023. arXiv: 2306.12234.

[13] EPTA Collaboration et al. "The second data release from the European Pulsar Timing Array. III. Search for gravitational wave signals". 2023. arXiv: 2306.16214.

[14] EPTA Collaboration et al. "The second data release from the European Pulsar Timing Array. I. The dataset and timing analysis". 2023. arXiv: 2306.16224.

[15] Clemente Smarra et al. "Second Data Release from the European Pulsar Timing Array: Challenging the Ultralight Dark Matter Paradigm". 2023. arXiv: 2306.16228.

[16] EPTA Collaboration et al. "The second data release from the European Pulsar Timing Array. II. Customised pulsar noise models for spatially correlated gravitational waves". 2023. arXiv: 2306.16225.

[17] Bhal Chandra Joshi et al. "Nanohertz gravitational wave astronomy during SKA era: An InPTA perspective". 2022. arXiv: 2207.06461.

[18] Pratik Tarafdar et al. "The Indian Pulsar Timing Array: First data release". 2022. arXiv: 2206.09289.

[19] K. Nobleson et al. "Low-frequency wideband timing of InPTA pulsars observed with the uGMRT". 2022. arXiv: 2112.06908.

[20] Jaikhomba Singha et al. "Evidence for profile changes in PSR J1713+0747 using the uGMRT". 2021. arXiv: 2107.04607.

[21] M. A. Krishnakumar et al. "High precision measurements of interstellar dispersion measure with the upgraded GMRT". 2021. arXiv: 2101.05334.

[22] Jaikhomba Singha et al. "Low frequency view of profile change event in PSR J1713+0747 with the uGMRT". 2021.

[23] Abhimanyu Susobhanan, Yogesh Maan, Bhal Chandra Joshi, T. Prabu, Shantanu Desai, K. Nobleson, **Sai Chaitanya Susarla**, et al. "pinta: The uGMRT data processing pipeline for the Indian Pulsar Timing Array". In: *PASA* 38, e017 (Apr. 2021).
Published in PASA. e017. DOI: 10.1017/pasa.2021.12. arXiv: 2007.02930 [astro-ph.IM].

Conference proceedings (Recent first):

[24] **Sai Chaitanya Susarla** & European Pulsar Timing Array (EPTA) Team. "Exploring the Effects of Solar Wind on Pulsars using the Low Frequency Array (LOFAR) Telescope". In: *APS April Meeting Abstracts*. Vol. 2023. APS Meeting Abstracts.
Conference Abstract for APS meeting 2023. Jan. 2023, T13.003, T13.003.

Computational Skills

- Proficient in: C++, Python and LaTeX.
- Working knowledge of: MATLAB, Mathematica
- Proficient in pulsar analysing softwares such as Tempo2, Sigproc, PSRchive, PINT, ENTERPRISE and LIBSTEMPO.
- Proficient in machine learning applications like the SBI software.

Supervisory and Teaching experience

- **Lab leader in Physics 2nd year undergraduate labs**
  University of Galway *Sep 2024–April 2025*

- **Supervised a summer project student**
  University of Galway *July 2024–Aug 2024*
  Title: Single pulses of the Crab pulsar using Irish-LOFAR station.
- **Lab leader in Physics 2nd year undergraduate labs**
  University of Galway *Sep 2023–Apr 2024*
- **Lab leader in Physics 1st year undergraduate labs**
  University of Galway *Sep 2021–Dec 2021*

Telescope proposals and Observing experience

- **LC51_050: PILLARs project - scintillation observations** **Allocated 23 hours**
  uGMRT observations *Oct 2026*
  Co observer
- **LC20_029: Tracing the Heliosphere with pulsars** **Allocated 24.5 hours**
  LOFAR Core proposal *June 2023*
  PI of the project.
- **LC18_016: Tracing the Heliosphere with pulsars** **Allocated 10.5 hours**
  LOFAR Core proposal *June 2022*
  PI of the project.
- **Chief observer handling the observations and processing of data**
  Irish-LOFAR station *Sep 2021–Aug 2025*
- **Scheduling and processing of observations.**
  German Long Wavelength LOFAR stations *Sep 2023–Jan 2025*
- **42_020: Star-planet interactions in the AU Mic planetary system**
  uGMRT observations *May 2022*
  Co-observer
- **Indian Pulsar Timing Array project**
  uGMRT observations *April 2019–Sept 2021*
  Co-observer

Memberships of Collaborations

- Square Kilometer Array pulsar Working Group — Aug 2024–Present
- LOFAR pulsar Working Group — Feb 2022–Present
- European Pulsar Timing Array — Oct 2021–Present
- International Pulsar Timing Array — May 2020–Present
- Indian Pulsar Timing Array — May 2018–Dec 2021

Known Languages

Telugu: Native
Hindi: Native
English: Fluent
Marathi: Basic

Other details

- Participated in outreach activity for students of Trinity College Dublin Space society at Birr, Ireland showing them the Irish-LOFAR telescope.
- Conducted Public Outreach for students in nearby schools in Thiruvananthapuram, India to spread awareness about Pure Sciences in March 2019.
- Represented Indian Institute of Science Education and Research (IISER) Thiruvananthapuram at the Inter IISER Sports Meet 2017 in Badminton
- Avid trekker and mountaineer; completed some of the toughest treks in the Western Ghats of India.
- Achieved a rank of ∼7000 in **Joint Entrance Examination Advanced 2016**
- Member of Sports council of IISER Thiruvananthapuram.